

复数及其四则运算

Complex Numbers and Their Four Fundamental Operations

选择题 (Multiple-Choice Questions)

1. 下面 4 种说法中正确的是()

Which of the following 4 statements is correct?

A. 如果实数 $a = b$, 则 $a - b + (a + b)i$ 是纯虚数; (If real numbers $a = b$, then $a - b + (a + b)i$ is a pure imaginary number;)

B. 模相等的两个复数是共轭复数; (Two complex numbers with equal moduli are conjugate complex numbers;)

C. 如果 z 是纯虚数, 则 $z \neq \bar{z}$; (If z is a pure imaginary number, then $z \neq \bar{z}$;))

D. 任何数的偶次幂不小于 0. (The even power of any number is not less than 0.)

2. 下面给出 4 个不等式, 其中正确的是()

The following 4 inequalities are given, which one is correct?

A. $3i > 2i$

B. $|2 + 3i| > |1 - 4i|$

C. $|2 - i| > 2i^4$

D. $i^2 > -i$

3. $a = 0$ 是复数 $a + bi$ ($a, b \in R$) 为纯虚数的()

$a = 0$ is a(n) () for the complex number $a + bi$ (where $a, b \in R$) to be a pure imaginary number.

A. 充分不必要条件 (Sufficient but not necessary condition)

B. 必要不充分条件 (Necessary but not sufficient condition)

C. 充分必要条件 (Sufficient and necessary condition)

D. 既不充分又不必要条件 (Neither sufficient nor necessary condition)

4. 复数 $z = \sin \frac{\pi}{3} + \cos \frac{\pi}{6} i$, 则 $|z|^2 = (\quad)$

For the complex number $z = \sin \frac{\pi}{3} + \cos \frac{\pi}{6} i$, then $|z|^2 = (\quad)$

A. $\frac{3}{4}$

B. $\frac{\sqrt{6}}{2}$

C. $\frac{3}{2}$

D. 1

5. 已知复数 z 满足 $|z|^2 - 2|z| - 3 = 0$, 则复数 z 的对应点的轨迹方程是()

Given that the complex number z satisfies $|z|^2 - 2|z| - 3 = 0$, the trajectory equation of the corresponding point of the complex number z is ()

A. 1 个圆 (One circle)

B. 线段 (A line segment)

C. 2 个点 (Two points)

D. 2 个圆 (Two circles)

6. 方程 $z^2 - 5|z| + 6 = 0$ 的复数集内解的个数是()

The number of solutions of the equation $z^2 - 5|z| + 6 = 0$ in the set of complex numbers is ()

- A. 4
- B. 6
- C. 8
- D. 2

7. 若虚数 $z = a + bi$ ($a, b \in \mathbb{R}$) 则 $|z^2|$ 、 $|z|^2$ 、 z^2 的关系是()

If the imaginary number $z = a + bi$ (where $a, b \in \mathbb{R}$), then the relationship between $|z^2|$, $|z|^2$, and z^2 is ()

- A. 互不相等 (Mutually unequal)
- B. $|z^2| = |z|^2 \neq z^2$
- C. $|z^2| \neq |z|^2 = z^2$
- D. $|z^2| = |z|^2 = z^2$

8. 设 $z = (2t^2 + 5t - 3) + (t^2 + 2t + 2)i$ ($t \in \mathbb{R}$), 则下列结论正确的是()

Let $z = (2t^2 + 5t - 3) + (t^2 + 2t + 2)i$ (where $t \in \mathbb{R}$), then which of the following conclusions is correct?

- A. z 对应的点在第一象限 (The corresponding point of z is in the first quadrant)
- B. z 一定不是纯虚数 (z is definitely not a pure imaginary number)
- C. z 对应的点在实轴下方 (The corresponding point of z is below the real axis)
- D. z 一定不是实数 (z is definitely not a real number)

9. a, b 都是复数, 下列命题中, 错误的是()

Both a and b are complex numbers. Which of the following propositions is incorrect?

- A. 若 $a = b$, 则 $\bar{a} = \bar{b}$ (If $a = b$, then $\bar{a} = \bar{b}$)
- B. $\sqrt{|a|^2} = |a|$
- C. 若 $|a| = |b|$, 则 $a = \pm b$ (If $|a| = |b|$, then $a = \pm b$)
- D. 如果 $a = \bar{b}$, 则 $a + b$ 为实数 (If $a = \bar{b}$, then $a + b$ is a real number)

10. 若复数 $(a + 1) + (a - 1)i \neq 0$, 则 a 的值是()

If the complex number $(a + 1) + (a - 1)i \neq 0$, then the value of a is ()

- A. 不等于 1 的实数 (A real number not equal to 1)
- B. 不等于 -1 的实数 (A real number not equal to -1)
- C. 不等于 ± 1 的实数 (A real number not equal to ± 1)
- D. 任意实数 (Any real number)

11. 已知 $z = \log_{\sin 1} \tan 1 + i \log_{\tan 1} \sin 1$, 那么 z 对应点所在象限是()

Given $z = \log_{\sin 1} \tan 1 + i \log_{\tan 1} \sin 1$, the quadrant where the corresponding point of z lies is ()

- A. 第一象限 (First quadrant)
- B. 第二象限 (Second quadrant)
- C. 第三象限 (Third quadrant)
- D. 第四象限 (Fourth quadrant)

12. 设 $z \in \mathbb{C}$, 若 $M = z|(z - 1)^2| = |z - 1|^2$, 则 M 是()

Let $z \in \mathbb{C}$. If $M = z|(z-1)^2| = |z-1|^2$, then M is ()

- A. $M =$ 实数 ($M =$ real numbers)
- B. $M =$ 虚数 ($M =$ imaginary numbers)
- C. 实数 $\subset M \subset$ 复数 (real numbers $\subset M \subset$ complex numbers)
- D. $M =$ 复数 ($M =$ complex numbers)

13. 已知虚数 $(x-2) + yi$ ($x, y \in \mathbb{R}$) 的模为 $\sqrt{3}$, 则 $\frac{y}{x}$ 的最大值是()

Given that the modulus of the imaginary number $(x-2) + yi$ (where $x, y \in \mathbb{R}$) is $\sqrt{3}$, the maximum value of $\frac{y}{x}$ is ()

- A. $\frac{\sqrt{3}}{2}$
- B. $\frac{\sqrt{3}}{3}$
- C. $\frac{1}{2}$
- D. $\sqrt{3}$

14. 已知复数 $|z| = \frac{1}{2}$, 则 $|z^2 - z + \frac{1}{4}|$ 的最大值是()

Given that the complex number $|z| = \frac{1}{2}$, the maximum value of $|z^2 - z + \frac{1}{4}|$ is ()

- A. 2
- B. 1
- C. $\frac{3}{4}$
- D. $\frac{1}{2}$

15. 满足条件 $x|x^2 + 1| = 0, x \in \mathbb{R} \subsetneq M \subseteq m || \log_3 m + 4i| = 5, m \in \mathbb{R}^+$ 的所有集合 M 的个数是()

The number of all sets M satisfying the condition $x|x^2 + 1| = 0, x \in \mathbb{R} \subsetneq M \subseteq m || \log_3 m + 4i| = 5, m \in \mathbb{R}^+$ is ()

- A. 4
- B. 3
- C. 2
- D. 1

16. 若 z 是复数且 $|z + 5 + 12i| = 2$, 则 $|z|$ 最小值是()

If z is a complex number and $|z + 5 + 12i| = 2$, the minimum value of $|z|$ is ()

- A. 8
- B. 9
- C. 10
- D. 11

17. 若复数 z 满足 $|z + 3 - \sqrt{3}i| = \sqrt{3}$, 则 $|z|$ 的最大值为 a , $|z|$ 的最小值为 b , 则 $a + b$ 的值是()

If the complex number z satisfies $|z + 3 - \sqrt{3}i| = \sqrt{3}$, let a be the maximum value of $|z|$ and b be the minimum value of $|z|$, then the value of $a + b$ is ()

- A. $2\sqrt{3}$

B. $3\sqrt{3}$

C. $4\sqrt{3}$

D. $5\sqrt{3}$

18. 设 z 满足 $|z+i|+|z-i|=2$, 那么 $|z+i+1|$ 的最小值是()

Let z satisfy $|z+i|+|z-i|=2$, then the minimum value of $|z+i+1|$ is ()

A. 1

B. $\frac{1}{2}$

C. 2

D. 3

19. 复平面内, 曲线 $|z-1+i|=1$ 关于直线 $y=x$ 的对称曲线方程为()

In the complex plane, the equation of the symmetric curve of the curve $|z-1+i|=1$ with respect to the line $y=x$ is ()

A. $|z+1+i|=1$

B. $|\bar{z}+1+i|=1$

C. $|z-1-i|=1$

D. $|\bar{z}-1-i|=1$

20. 集合 $P = \{z \mid |z+2|+|z-2|=6, z \in \mathbb{C}\}$, $Q = \{z \mid |z+1|=1, z \in \mathbb{C}\}$, 则 P 与 Q 的关系是()

Let set $P = \{z \mid |z+2|+|z-2|=6, z \in \mathbb{C}\}$ and $Q = \{z \mid |z+1|=1, z \in \mathbb{C}\}$, then the relationship between P and Q is ()

A. $P \subset Q$

B. $Q \subset P$

C. $P = Q$

D. $P \cap Q = \emptyset$

21. 复平面内, 两点 M, N 所对应的非零复数 α, β , 满足 $\alpha^2 + \beta^2 = 0$, 则 $\triangle OMN$ (O 为坐标原点) 是()

In the complex plane, two points M and N correspond to non-zero complex numbers α and β respectively, satisfying $\alpha^2 + \beta^2 = 0$. Then $\triangle OMN$ (where O is the origin of coordinates) is ()

A. 等腰直角三角形 (Isosceles right triangle)

B. 等边三角形 (Equilateral triangle)

C. 非直角的等腰三角形 (Isosceles triangle that is not right-angled)

D. 非等腰的直角三角形 (Right triangle that is not isosceles)

22. 已知关于 x 的方程 $x^2 + x + p = 0$ 的两虚根为 x_1 和 x_2 , 且 $|x_1 - x_2| = 3$, 则实数 p 的值为()

Given that the two imaginary roots of the equation $x^2 + x + p = 0$ about x are x_1 and x_2 , and $|x_1 - x_2| = 3$, the value of the real number p is ()

A. -1

B. 1

C. $-\frac{5}{2}$

D. $\frac{5}{2}$

23. 设复数 z_1, z_2 满足 $|z_1| = |z_2| = 2$, $|z_1 + z_2| = 2\sqrt{3}$, 则 $|z_1 - z_2| = ()$

Let the complex numbers z_1, z_2 satisfy $|z_1| = |z_2| = 2$ and $|z_1 + z_2| = 2\sqrt{3}$, then $|z_1 - z_2| = ()$

A. 1

B. $\sqrt{3}$

C. 2

D. 3

24. 已知 i 是虚数单位, 则能使 $(n + i)^4$ 成为整数的整数 n 的个数是()

Given that i is the imaginary unit, the number of integers n that can make $(n + i)^4$ an integer is ()

A. 2

B. 3

C. 4

D. 无数个 (Countless)

25. 已知 $f(\bar{z} + i) = z + 2\bar{z} - 2i$, 则 $f(i^5)$ 等于()

Given $f(\bar{z} + i) = z + 2\bar{z} - 2i$, then $f(i^5)$ equals ()

A. $1 + i$

B. $1 - i$

C. $2i$

D. $-2i$

26. 已知 $f(n) = i^n - i^{-n}$ ($i^2 = -1$, $n \in N$), 集合 $f(n)$ 的元素个数是()

Given $f(n) = i^n - i^{-n}$ (where $i^2 = -1$, $n \in N$), the number of elements in the set $f(n)$ is ()

A. 2

B. 3

C. 4

D. 无数个 (Countless)

27. $i + i^2 + i^3 + \dots + i^{2000}$ 等于()

$i + i^2 + i^3 + \dots + i^{2000}$ equals ()

A. 0

B. -1

C. 1

D. i

28. $f(z) = 1 - \bar{z}$, $z_1 = 2 + 3i$, $z_2 = 5 - i$, 则 $\overline{f(z_1 - z_2)}$ 等于()

$f(z) = 1 - \bar{z}$, $z_1 = 2 + 3i$, $z_2 = 5 - i$, then $\overline{f(z_1 - z_2)}$ equals ()

A. $-3 - 4i$

B. $3 - 4i$

C. $4 - 4i$

D. $4 + 4i$

29. 复数 α, β 分别对应于复平面内的点 P, Q , O 为原点, 若 $\alpha^2 - 2\alpha\beta + 4\beta^2 = 0$, 则 $\triangle POQ$ 是

()

The complex numbers α, β correspond to points P, Q in the complex plane respectively, and O is the origin. If $\alpha^2 - 2\alpha\beta + 4\beta^2 = 0$, then $\triangle POQ$ is ()

- A. 等腰直角三角形 (Isosceles right triangle)
- B. 等边三角形 (Equilateral triangle)
- C. 一锐角为 60° 的直角三角形 (Right triangle with an acute angle of 60°)
- D. 顶角为 30° 的等腰三角形 (Isosceles triangle with a vertex angle of 30°)

30. 复数 $\left(\frac{1-i}{1+i}\right)^{13}$ 的值等于()

The value of the complex number $\left(\frac{1-i}{1+i}\right)^{13}$ is equal to ()

- A. $\frac{\sqrt{2}}{2}$
- B. $\sqrt{2}$
- C. i
- D. $-i$

31. 设 $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$, 那么 $1 + \omega + \omega^2 + \cdots + \omega^{13}$ 的值是()

Let $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$, then the value of $1 + \omega + \omega^2 + \cdots + \omega^{13}$ is ()

- A. 0
- B. $\frac{\sqrt{3}}{2} + \frac{1}{2}i$
- C. $-\frac{1}{2} - \frac{\sqrt{3}}{2}i$
- D. $\frac{1}{2} + \frac{\sqrt{3}}{2}i$

32. 设复数 z 满足 $z + \frac{4}{z}$ 为实数, 且 $\arg(z - 2) = \frac{3\pi}{4}$, 求复数 $z = (\quad)$

Let the complex number z satisfy that $z + \frac{4}{z}$ is a real number and $\arg(z - 2) = \frac{3\pi}{4}$, then the complex number $z = (\quad)$

- A. $2i$
- B. $1 + i$
- C. $-1 + i$
- D. $1 - i$

33. 已知 z_1, z_2 是两个非 0 复数, 且 $|z_1 + z_2| = |z_1 - z_2|$, 则 $\left(\frac{z_1}{z_2}\right)^2$ 是()

Given that z_1, z_2 are two non-zero complex numbers and $|z_1 + z_2| = |z_1 - z_2|$, then $\left(\frac{z_1}{z_2}\right)^2$ is ()

- A. 正数 (Positive number)
- B. 负数 (Negative number)
- C. 零 (Zero)
- D. 虚数 (Imaginary number)